

Sept. 28th.

Outline:

- ① Point estimation, order statistics, Gamma dist
- ② Confidence interval.

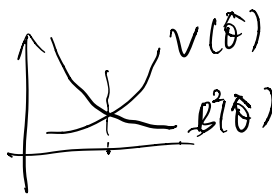
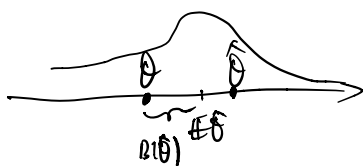
8.1 Using the identity

$$(\hat{\theta} - \theta) = [\hat{\theta} - E(\hat{\theta})] + [E(\hat{\theta}) - \theta] = [\hat{\theta} - E(\hat{\theta})] + B(\hat{\theta}),$$

show that

$$\text{MSE}(\hat{\theta}) = E[(\hat{\theta} - \theta)^2] = V(\hat{\theta}) + (B(\hat{\theta}))^2.$$

$$\begin{aligned} E(\hat{\theta} - \theta)^2 &= E(\hat{\theta} - E\hat{\theta} + E\hat{\theta} - \theta)^2 \\ &= E[(\hat{\theta} - E\hat{\theta})^2 + 2(\hat{\theta} - E\hat{\theta})(E\hat{\theta} - \theta) + (E\hat{\theta} - \theta)^2] \\ &= E(\hat{\theta} - E\hat{\theta})^2 + 2\underbrace{E(\hat{\theta} - E\hat{\theta})}_{=0} \underbrace{(E\hat{\theta} - \theta)}_{=0} + E(E\hat{\theta} - \theta)^2 \\ &= \text{Var } \hat{\theta} + B^2(\hat{\theta}) \end{aligned}$$



8.8 Suppose that Y_1, Y_2, Y_3 denote a random sample from an exponential distribution with density function

$$f(y) = \begin{cases} \left(\frac{1}{\theta}\right) e^{-y/\theta}, & y > 0, \\ 0, & \text{elsewhere.} \end{cases}$$

Consider the following five estimators of θ :

$$\hat{\theta}_1 = Y_1, \quad \hat{\theta}_2 = \frac{Y_1 + Y_2}{2}, \quad \hat{\theta}_3 = \frac{Y_1 + 2Y_2}{3}, \quad \hat{\theta}_4 = \min(Y_1, Y_2, Y_3), \quad \hat{\theta}_5 = \bar{Y}.$$

- a Which of these estimators are unbiased?
- b Among the unbiased estimators, which has the smallest variance?

Expo(θ) has pdf $f(x) = \theta e^{-\theta x}$, $x \geq 0$.

$Y_1, Y_2, Y_3 \sim \text{Expo}(\frac{1}{\theta})$. If $X \sim \text{Expo}(\theta)$, $\mathbb{E}X = \frac{1}{\theta}$.

$$\mathbb{E}Y_1 = \mathbb{E}Y_2 = \mathbb{E}Y_3 = \theta.$$

$$\text{Var} X = \frac{1}{\theta^2}.$$

$$\text{Var} Y_1 = \theta^2.$$

a. $\hat{\theta}_1 = Y_1$, $\mathbb{E}Y_1 = \theta$.

$$\hat{\theta}_2 = \frac{Y_1 + Y_2}{2}, \quad \mathbb{E} \frac{Y_1 + Y_2}{2} = \frac{1}{2} \mathbb{E}(Y_1 + Y_2) = \frac{1}{2}(\mathbb{E}Y_1 + \mathbb{E}Y_2) = \theta$$

$$\hat{\theta}_3 = \frac{Y_1 + Y_2 + Y_3}{3}$$

$$\hat{\theta}_5 = \bar{Y}$$

$$\hat{\theta}_4 = \min(Y_1, Y_2, Y_3) = Y_{(1)}$$

$$X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} X, \quad X_{(1)} \leq X_{(2)} \leq \dots \leq X_{(n)}$$

pdf of X is $f(x)$, cdf is $F(x)$ $f(x) = F'(x)$

$$\begin{aligned} P(X_{(1)} \geq x) &= P(X_1, \dots, X_n \geq x) = \prod_{i=1}^n P(X_i \geq x) \\ &= (1 - F(x))^n \\ &= 1 - F_{X_{(1)}}(x) \end{aligned}$$

$$P(X_{(n)} \leq x) = P(X_1, \dots, X_n \leq x) = F(x)^n.$$

$$\hat{\theta}_4 = Y_{(1)} \quad F_Y(y) = e^{-\frac{y}{\theta}}$$

$$\hat{\theta}_4 = \min(Y_1, Y_2, Y_3) \geq Y_1$$

$$F_{\hat{\theta}_4}(y) = 1 - (1 - F_Y(y))^3$$

$$\mathbb{E}\hat{\theta}_4 > \mathbb{E}Y_1 = \theta.$$

$$f_{\hat{\theta}_4}(y) = 3f_Y(y)(1 - F_Y(y))^2.$$

$$= 3 \frac{1}{\theta} e^{-\frac{y}{\theta}} (1 - e^{-\frac{y}{\theta}})^2.$$

***8.16** Suppose that $Y_1, Y_2, \dots, Y_n$ constitute a random sample from a normal distribution with parameters μ and σ^2 .¹

a Show that $S = \sqrt{S^2}$ is a biased estimator of σ . [Hint: Recall the distribution of $(n-1)S^2/\sigma^2$ and the result given in Exercise 4.112.]

b Adjust S to form an unbiased estimator of σ .

c Find an unbiased estimator of $\mu - z_\alpha \sigma$, the point that cuts off a lower-tail area of α under this normal curve.

$$X \sim \text{Gamma}(\alpha, \beta) \quad \text{pdf} \quad f(x) = \frac{1}{\Gamma(\alpha) \beta^\alpha} \underbrace{x^{\alpha-1} e^{-\frac{x}{\beta}}}_{\Gamma(\alpha) = \int_0^\infty x^{\alpha-1} e^{-x} dx}. \quad \int_0^\infty f(x) dx = 1$$

$$\mathbb{E}X = \alpha\beta, \quad \text{Var} X = \alpha\beta^2.$$

$$\text{Expo}(\theta) \sim \text{Gamma}(1, \frac{1}{\theta})$$

$$Y \sim \underline{N(\mu, \sigma^2)}$$

$$\underline{\chi_n^2 \sim \text{Gamma}(\frac{n}{2}, 2)}$$

$$a. \frac{(n-1)S^2}{\sigma^2} \sim \chi_{n-1}^2 \sim \text{Gamma}(\frac{n-1}{2}, 2) \sim X$$

$$\begin{aligned} \mathbb{E} S &= \mathbb{E} (S^2)^{\frac{1}{2}} = \mathbb{E} \left(\frac{(n-1)S^2}{\sigma^2} \right)^{\frac{1}{2}} \cdot \left(\frac{n-1}{\sigma^2} \right)^{-\frac{1}{2}} \\ &= \frac{\sigma}{\sqrt{n-1}} \mathbb{E} X^{\frac{1}{2}} \\ &= \frac{\sigma}{\sqrt{n-1}} \int_0^\infty x^{\frac{1}{2}} \cdot \frac{1}{\Gamma(\frac{n-1}{2}) 2^{\frac{n-1}{2}}} x^{\frac{n-1}{2}-1} e^{-\frac{x}{2}} dx \\ &= \frac{\sigma}{\sqrt{n-1}} \cdot \frac{\Gamma(\frac{n-1}{2} + \frac{1}{2}) 2^{\frac{n-1}{2} + \frac{1}{2}}}{\Gamma(\frac{n-1}{2}) 2^{\frac{n-1}{2}}} \\ &= \boxed{\frac{\Gamma(\frac{n-1}{2} + \frac{1}{2})}{\Gamma(\frac{n-1}{2})} \cdot \sqrt{\frac{2}{n-1}}} \sigma \rightarrow C_n. \end{aligned}$$

$$b. \mathbb{E} \frac{1}{C_n} S = \sigma.$$

$$c. \mu - z_{\alpha} \frac{\sigma}{\sqrt{n}}, \quad \bar{Y} - z_{\alpha} \frac{S}{\sqrt{n}}$$

8.43 Let $Y_1, Y_2, \dots, Y_n$ denote a random sample of size n from a population with a uniform distribution on the interval $(0, \theta)$. Let $Y_{(n)} = \max(Y_1, Y_2, \dots, Y_n)$ and $U = (1/\theta)Y_{(n)}$.

a Show that U has distribution function

$$F_U(u) = \begin{cases} 0, & u < 0, \\ u^n, & 0 \leq u \leq 1, \\ 1, & u > 1. \end{cases}$$

b Because the distribution of U does not depend on θ , U is a pivotal quantity. Find a 95% lower confidence bound for θ .

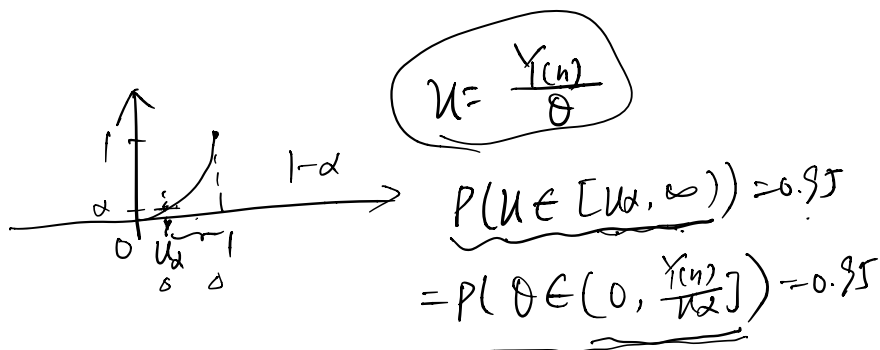
$$a. F_Y(y) = \frac{y}{\theta}, \quad f_Y(y) = \frac{1}{\theta} \mathbb{1}_{\{y \leq \theta\}}.$$

$$F_{Y_{(n)}}(y) = F_Y^n(y) = \left(\frac{y}{\theta}\right)^n, \quad 0 \leq y \leq \theta$$

$$U = \frac{Y_{(n)}}{\theta}, \quad F(U \leq u) = F\left(\frac{Y_{(n)}}{\theta} \leq u\right) \\ = F_{Y_{(n)}}(u\theta) = u^n, \quad 0 \leq u \leq 1$$

$$b. P\left(\frac{Y_{(n)}}{\theta} \in [\underline{u}_\alpha, \infty)\right) = 0.95$$

$$= P\left(\theta \in \left(0, \frac{Y_{(n)}}{u_\alpha}\right]\right) = 0.95$$



8.46

Refer to Example 8.4 and suppose that Y is a single observation from an exponential distribution with mean θ .

- a Use the method of moment-generating functions to show that $2Y/\theta$ is a pivotal quantity and has a χ^2 distribution with 2 df.
- b Use the pivotal quantity $2Y/\theta$ to derive a 90% confidence interval for θ .